

Task Division — Talk-to-my-Data

Rotation across all 4 pillars: everyone takes a different sub-task each pillar, so by Stage 3 everyone has touched data, AI, backend, and frontend. Tasks marked Required come directly from Artefact's brief — don't skip these. Tasks marked Stretch are what makes the project stand out, but are the first things to cut if the timeline gets tight.

Note: Stage numbering and pillar numbering are separate — Stage 0's tasks are 0.1-0.4, Stage 1's are 1.1-1.3, and each pillar's tasks restart at .1 within that pillar.

Stage 0 — Prerequisites ☒ COMPLETE

Sub-task	Owner	Priority	Deadline
0.1 Cloud Setup (GCP project, CLI, service decisions) ☒ DONE	Raed	☒ Done	Closed
0.2 Environment Setup (Python, venv, IDE, .gitignore) ☒ DONE	Muhammed	☒ Done	Closed
0.3 Architecture Sketch v1 (Excalidraw) ☒ DONE	Amir	☒ Done	Closed
0.4 Dataset & KPI Draft ☒ DONE	Fares	☒ Done	Closed

Gate cleared: GCP project verified working end-to-end, environment documented, architecture sketch presented to Mark + Quinten, dataset downloaded and KPIs defined.

Stage 1 — Contracts & Skeleton ☒ COMPLETE

Sub-task	Owner	Priority	Deadline
1.1 Write schema.md ☒ DONE	All (together)	☒ Done	Closed
1.2 Write api.md ☒ DONE	All (together)	☒ Done	Closed
1.3 Repo Skeleton (/src, /mock_data, /docs) ☒ DONE	All (together)	☒ Done	Closed

Gate cleared: both contract files committed and rendering correctly, folder skeleton matches the plan.

Stage 2 — Core Build

Pillar 1 — Cloud & Data Platform

Sub-task	Owner	Priority	Deadline
1.1 Retail Data Cleaning & ETL ☒ DONE	Raed	Complete ☒	Sat, Sep 19
1.2 BigQuery Data Warehouse Setup ☒ DONE	Fares	Complete ☒	Sat, Sep 19

Gate: real data queryable in BigQuery by everyone.

Note: 1.3 (IAM & Security) and 1.4 (Docker & Cloud Run) moved out of this pillar -- see reasoning below.

Pillar 2 — LLM & Text-to-SQL

Sub-task	Owner	Priority	Deadline
2.1 Dynamic Schema Injection	Raed	Required — core ask	Sat, Sep 26
2.2 Few-Shot Prompting & Domain Lexicon	Amir	Stretch	Sat, Sep 26
2.3 Self-Healing Query Feedback Loop	Muhammed	Stretch	Sat, Sep 26
2.4 Ambiguity Resolution & Clarification	Fares	Stretch	Sat, Sep 26

Gate: a plain-English question reliably produces correct SQL against real data.

Pillar 3 — Backend & Analytics Engine

Sub-task	Owner	Priority	Deadline
3.1 FastAPI Core Service Layer	Amir	Required	Sat, Sep 26
3.2 SQL Validation & Safety Guardrails	Muhammed	Required	Sat, Sep 26
3.3 Retail KPI Computation Engine	Raed	Required — brief asks for 2+ KPIs	Sat, Sep 26
3.4 Multi-Modal Response Formatter	Fares	Required	Sat, Sep 26

Gate: /ask endpoint works end-to-end, callable directly, no UI needed yet.

Pillar 4 — Frontend UI, Benchmark & Delivery

Sub-task	Owner	Priority	Deadline
4.1 Streamlit Conversational Interface	Muhammed	Required	Sat, Oct 3
4.2 Dynamic Data Visualizations	Raed	Stretch	Sat, Oct 3
4.3 Golden Benchmark Evaluation Suite	Fares	Stretch (10-15 questions is fine)	Sat, Oct 3
4.4 Excalidraw Architecture & Client Deck	Amir	Required — named in the brief	Sat, Oct 3

Gate: a real question typed into the real UI returns a real, working answer.

Stage 3 — Integration

Sub-task	Owner	Priority	Deadline
Full Pipeline Integration (replace remaining mocks, test end-to-end)	All (together)	Required	Wed, Oct 7
1.4 Docker & Cloud Run Deployment (moved here from Pillar 1 -- nothing existed yet to containerize)	Muhammed	Required	Wed, Oct 7

Gate: the full system works start to finish, without anyone manually patching things together.

1.4 moved here from Pillar 1: deploying to Cloud Run only makes sense once there's a real frontend+backend to containerize -- doing it earlier would mean deploying nothing meaningful.

Stage 4 — Dry Run Prep

Sub-task	Owner	Priority	Deadline
Dry Run Deck + Rehearsal, presented to Mark + Quinten	All (together)	Required	Wed, Oct 7

Gate: feedback received, clear list of what to fix before the final.

Stage 5 — Final Polish & Deployment

Sub-task	Owner	Priority	Deadline
Full Cloud Run Deployment + Final Deck + Live Demo Rehearsal	All (together)	Required	Wed, Oct 7
1.3 IAM & Security Configuration (deferred from Pillar 1 -- see note)	Amir	Stretch	Wed, Oct 7

Gate: live demo works, deck is ready, everyone knows their speaking part.

1.3 deferred here from Pillar 1: tightening access mid-build risks blocking someone during the busiest weeks. Safer once the system is stable and access needs are fully known. Raed (and at least one other) always keep full Editor access as a safety net, regardless of this task.